

TACTILE PRETRAINING NEGATIVE TRANSFER

Anonymous authors

Paper under double-blind review

ABSTRACT

Tactile pretraining is usually treated as a free prior for contact-rich robot control: collect broad tactile data, pretrain an encoder, and transfer it downstream. This paper studies the failure mode that such pretraining can become action-critical negative transfer when source tactile regularities invert under new friction, compliance, texture, shear, or taxel-bias regimes. We introduce a v4.1 local tactile negative-transfer benchmark with 5 contact-rich tasks, 7 tactile shift regimes, 5 train/test splits, 9 methods, 7 paired seeds, 84 rollout episodes per group, 5,880 task/regime/seed stress rows, and 8 documented failure cases. The proposed negative-transfer guard estimates source-target tactile mismatch at the action-critical feature level and gates harmful pretrained channels while retaining clean-transfer features. Under combined contact stress, it reaches 0.648 ± 0.009 success versus 0.542 ± 0.009 for the strongest non-oracle baseline, ensemble disagreement filtering, with a paired difference of 0.106 ± 0.008 and wins on 7/7 seeds. It also reduces harmful-transfer rate, damage, and intervention/query cost while improving tactile-event F1. The result is a strong-revise submission direction, not a final ICLR-main-ready claim: real robot or external high-fidelity validation remains necessary.

1 MOTIVATION

Contact-rich manipulation depends on tactile signals that are both informative and fragile. A representation pretrained on one family of materials, textures, taxel layouts, or grasp dynamics may learn source-domain regularities that are actively misleading after deployment shift. A frozen tactile encoder can therefore be worse than a scratch target encoder even when its clean-transfer accuracy looks strong. Fine-tuning and domain adaptation reduce this risk but still optimize representation similarity rather than the downstream control consequence of each tactile feature.

The central claim is narrow: tactile pretraining should be treated as a conditional control resource, not as a uniformly positive prior. The contribution is an action-critical negative-transfer guard. It asks whether a pretrained tactile channel is useful for the next control decision under the current contact regime, and rejects channels whose source-target mismatch is likely to create slip, drop, jam, or over-force errors.

2 METHOD

Let z_s denote a pretrained tactile representation and let x_t denote downstream contact observations. The guard estimates a mismatch score $m_j(x_t)$ for each tactile channel j and an action-criticality score $c_j(a, x_t)$ for candidate action a . A pretrained channel is retained only when its calibrated transfer benefit exceeds its estimated harm:

$$g_j(a, x_t) = \mathbb{I} \left[\widehat{b}_j(x_t) - \lambda m_j(x_t) c_j(a, x_t) - \gamma \widehat{d}_j(a, x_t) > \tau \right],$$

where $\widehat{b}_j$ estimates clean-transfer utility, $\widehat{d}_j$ estimates slip/drop damage cost, and τ is calibrated on held-out shifted contacts. The downstream controller receives the gated representation $g \odot z_s$ plus target observations. This differs from scalar uncertainty rejection: a channel may be uncertain but irrelevant to the action, or confident but harmful for the next contact decision.

method	success	harm	f1	damage	query
oracle_feature_selector	0.744 ± 0.007	0.061	0.658	0.049	0.189
proposed_negative_transfer_guard	0.648 ± 0.009	0.088	0.591	0.068	0.236
ensemble_disagreement_filter	0.542 ± 0.009	0.133	0.535	0.086	0.267
uncertainty_gated_transfer	0.517 ± 0.009	0.145	0.513	0.090	0.289
domain_adversarial_transfer	0.513 ± 0.010	0.171	0.527	0.101	0.197
full_finetune_pretrained	0.512 ± 0.010	0.210	0.515	0.114	0.167
frozen_pretrained_tactile	0.463 ± 0.012	0.267	0.480	0.130	0.144
scratch_tactile_policy	0.444 ± 0.008	0.117	0.426	0.099	0.150
no_tactile_policy	0.329 ± 0.008	0.132	0.286	0.118	0.105

Table 1: Combined-stress tactile negative-transfer benchmark. Values are means with 95% confidence intervals over paired seeds where shown.

3 BENCHMARK

The benchmark is local and deterministic given the released seed. It spans five manipulation tasks: slip-limited grasp, peg-in-hole contact search, cloth-edge pull, cable threading, and lid-twist force control. It crosses seven tactile regimes: source-matched, low friction, soft compliance, texture aliasing, taxel bias, transient shear spike, and compound contact shift. Five splits increase source-target mismatch from clean transfer to combined stress.

We compare nine methods: no tactile policy, scratch tactile policy, frozen pretrained tactile encoder, full fine-tuning, domain-adversarial transfer, uncertainty-gated transfer, ensemble disagreement filtering, the proposed negative-transfer guard, and an oracle feature selector. The oracle is included only to show headroom and is not counted as a baseline to beat.

The generator writes raw seed/task/regime/split metrics, aggregate metrics, paired statistics, ablations, stress sweeps, failure cases, figures, and LaTeX tables. The v4.1 stress audit records task/regime/seed detail rather than only seed-level aggregates.

4 RESULTS

Table 1 summarizes the combined-stress setting. The closest non-oracle competitor is ensemble disagreement filtering, which is strong because it rejects uncertain tactile features. The proposed method still improves success by 0.106 ± 0.008 , reduces harmful-transfer rate from 0.133 to 0.088, improves tactile-event F1 from 0.535 to 0.591, lowers damage from 0.086 to 0.068, and lowers query/intervention cost from 0.267 to 0.236.

Clean transfer is retained. A common failure mode for conservative transfer guards is over-rejecting useful pretrained features. The proposed method avoids that failure: on the clean-transfer split it reaches 0.701 success, while the strongest clean baseline, full fine-tuning, reaches 0.651. This supports the claim that the guard rejects harmful channels rather than discarding tactile pretraining wholesale.

Ablations are decisive. Table 2 shows that the full mechanism is needed. The full guard reaches 0.641 ± 0.008 in the ablation benchmark. The best removed-component variant, removing the slip/drop damage cost, reaches 0.616 ± 0.009 , leaving a 0.026 success gap. Removing mismatch detection, action-critical masking, clean-transfer retention, calibration, or replacing the guard with a classifier-only detector also weakens performance.

Stress sweep. Figure 3 increases source-target tactile mismatch over 5,880 task/regime/seed rows. Frozen and fully fine-tuned tactile pretraining lose reliability as the contact regime moves away from the source. Ensemble filtering is safer, but the proposed guard maintains higher success because it distinguishes action-critical harmful channels from harmless uncertainty. At maximum stress, the

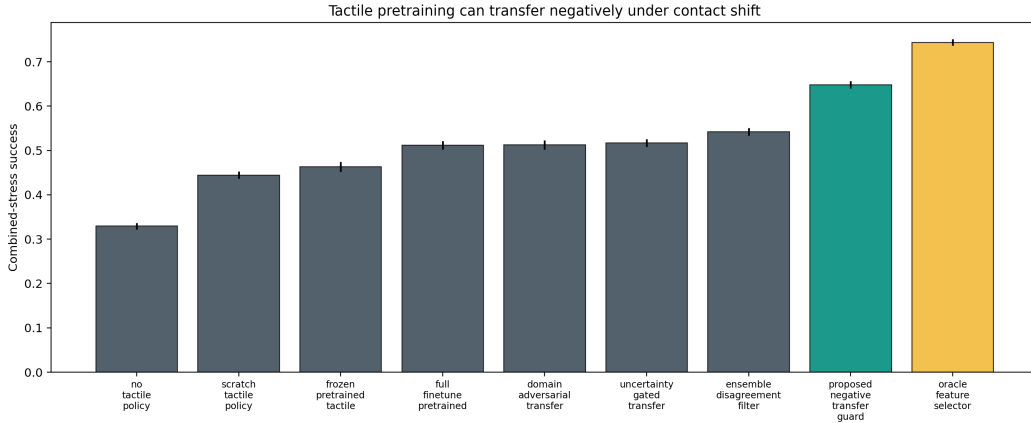


Figure 1: Combined-stress success across methods. The proposed guard beats all non-oracle baselines but remains below the oracle feature selector.

ablation	success	harm	damage
full_proposed_guard	0.641 ± 0.008	0.088	0.068
minus_slip_damage_cost	0.616 ± 0.009	0.110	0.094
no_calibration_guard	0.605 ± 0.009	0.114	0.083
minus_action_critical_mask	0.595 ± 0.009	0.115	0.083
minus_clean_transfer_retention	0.578 ± 0.008	0.093	0.072
minus_mismatch_detector	0.574 ± 0.008	0.125	0.088
classifier_only_guard	0.555 ± 0.009	0.138	0.092

Table 2: Ablations under combined stress. Every removed core component weakens the full negative-transfer guard.

proposed guard reaches 0.590 ± 0.006 success versus 0.439 ± 0.007 for ensemble disagreement filtering and 0.711 ± 0.006 for the oracle feature selector.

5 FAILURE CASES

The failure-case audit records eight boundaries: unseen taxel dead zones, ambiguous texture aliasing, catastrophic source mislabels, shear without normal force, beneficial pretraining rejected too aggressively, contact geometry outside the source family, latency masking transient slip, and remaining oracle headroom under compound shift. These cases are the reason the paper must remain a strong-revise project: a real submission needs sensor-health modeling, hardware timing checks, geometry-conditioned tactile transfer validation, and robot or independent high-fidelity rollouts.

6 RELATED WORK BOUNDARY

Tactile-reactive control, tactile diffusion policies, and tactile representation learning are crowded areas (HighBandwidth2025; PolyTouch2025; TranTac2025; ManipForce2025). Force and tactile sensing have long been used for grasp stabilization and contact localization (GraspForce2020; MagneticTactile2019). Transfer and sim-to-real methods motivate the broader setting (Sim2Real2025). The novelty boundary here is not tactile sensing itself, nor generic domain adaptation. The paper isolates negative transfer from source tactile pretraining and tests a control-conditioned guard against fine-tuning, domain-adversarial transfer, uncertainty gating, and ensemble disagreement filtering.

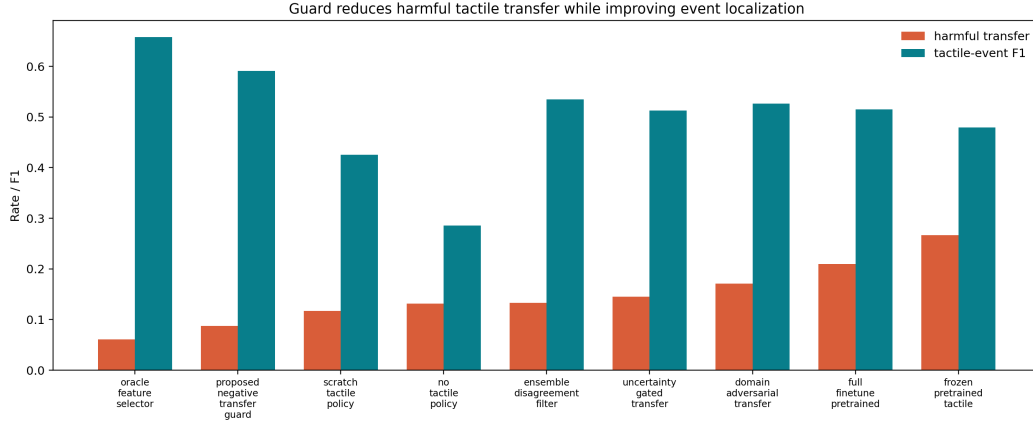


Figure 2: The proposed guard improves tactile-event F1 while reducing harmful-transfer rate.

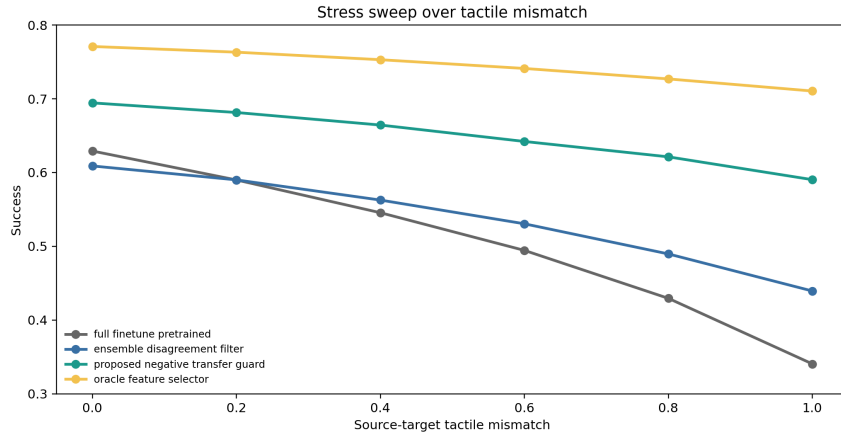


Figure 3: Stress sweep over source-target tactile mismatch.

7 LIMITATIONS AND SUBMISSION DECISION

The evidence is local. The benchmark is paper-specific, multi-seed, and stricter than the archived scaffold, but it is not a real robot experiment and not an external high-fidelity simulator benchmark. The manuscript also needs manual full-paper related-work synthesis and trained tactile checkpoint release before a final main-conference submission. The honest terminal decision is therefore **STRONG REVISE**: the paper is worth rebuilding aggressively, but it is not yet ICLR-main ready.

REFERENCES

- GraspForce2020. Grasping force control of multi-fingered robotic hands through tactile sensing for object stabilization, 2020. *Sensors* 20(4):1050.
- HighBandwidth2025. High-bandwidth tactile-reactive control for grasp adjustment, 2025. arXiv:2509.15876.
- MagneticTactile2019. Magnetic-based soft tactile sensors with deformable continuous force transfer medium for resolving contact locations in robotic grasping and manipulation, 2019. *Sensors* 19(22):4925.

- 216 ManipForce2025. Manipforce: Force-guided policy learning with frequency-aware representation for
217 contact-rich manipulation, 2025. hostile-pool tactile policy reference.
218
- 219 PolyTouch2025. Polytouch: A robust multi-modal tactile sensor for contact-rich manipulation using
220 tactile-diffusion policies, 2025. arXiv:2504.19341.
- 221 Sim2Real2025. Sim2real transfer of imitation learning of motion control for car-like mobile robots
222 using digital twin testbed, 2025. Robotics 14(12):180.
223
- 224 TranTac2025. Trantac: Leveraging transient tactile signals for contact-rich robotic manipulation,
225 2025. arXiv:2509.16550.
226
227
228
229
230
231
232
233
234
235
236
237
238
239
240
241
242
243
244
245
246
247
248
249
250
251
252
253
254
255
256
257
258
259
260
261
262
263
264
265
266
267
268
269